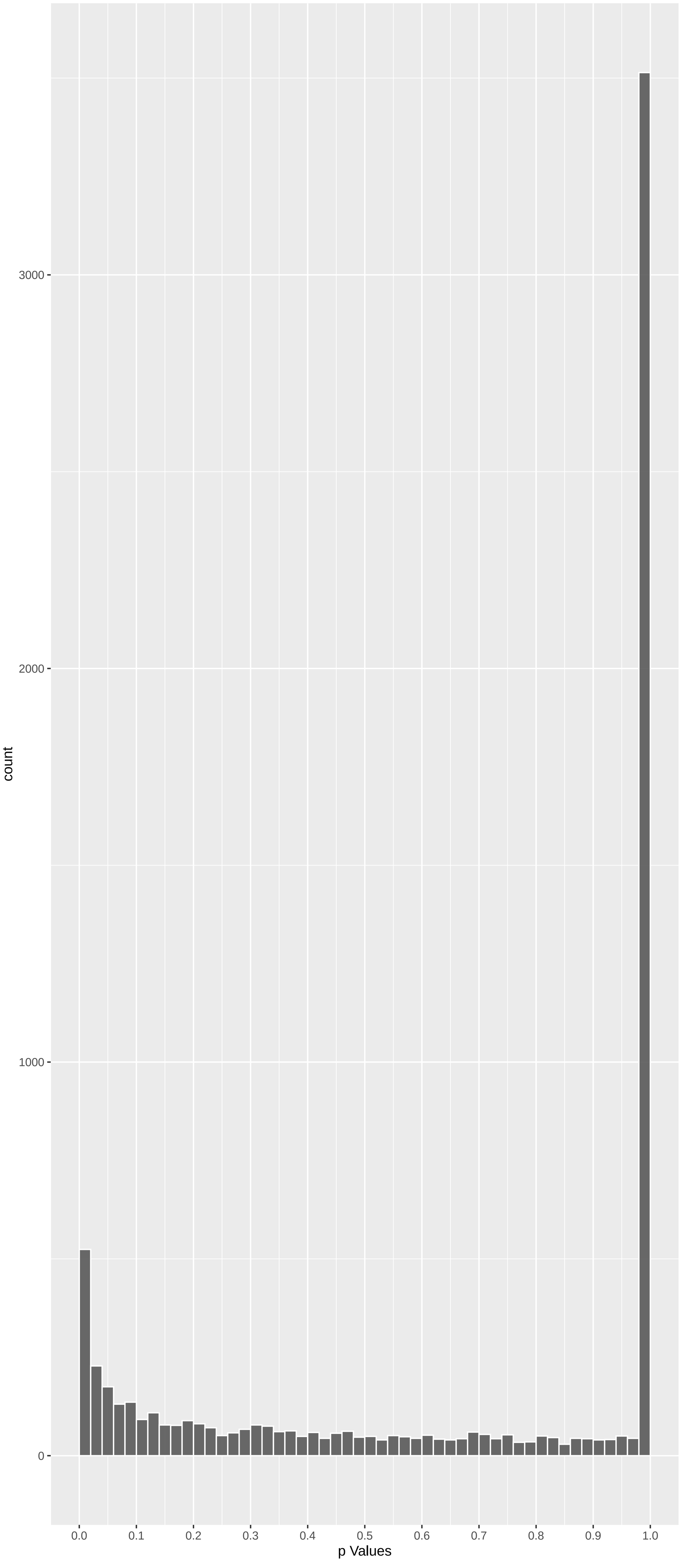
Top genes by P−value non−permulated

Gene	Rho	P	p.adj	qValueNoperm
LRPPRC	−5.814897	1.820133e−08	2.082e−04	1.000e+00
FBXO24	5.743811	2.777080e−08	2.082e−04	1.000e+00
METAP1D	−5.568879	7.691524e−08	3.845e−04	1.000e+00
CLCNKA	−5.462595	1.407671e−07	5.278e−04	1.000e+00
PIWIL1	5.370477	2.355855e−07	7.066e−04	1.000e+00
CATSPERE	−5.203486	5.867539e−07	1.338e−03	1.000e+00
TET1	−5.155633	7.583248e−07	1.338e−03	1.000e+00
THADA	−5.144860	8.031603e−07	1.338e−03	1.000e+00
TRANK1	5.180663	6.632962e−07	1.338e−03	1.000e+00
KIF18B	−5.040409	1.393614e−06	2.090e−03	1.000e+00
TMEM126A	−4.996415	1.752173e−06	2.190e−03	1.000e+00
CENPT	−5.008843	1.642747e−06	2.190e−03	1.000e+00
SLFN14	−4.891836	2.996988e−06	3.457e−03	1.000e+00
A2M	−4.786942	5.080259e−06	5.325e−03	1.000e+00
MYH13	4.764435	5.681512e−06	5.325e−03	1.000e+00
LPIN1	4.776008	5.364286e−06	5.325e−03	1.000e+00
MAN2B1	−4.720965	7.041859e−06	5.474e−03	1.000e+00
MUC20	−4.725378	6.890633e−06	5.474e−03	1.000e+00
TNNI3K	−4.698125	7.876804e−06	5.474e−03	1.000e+00
DST	−4.693967	8.038737e−06	5.474e−03	1.000e+00
KCNQ3	4.685089	8.395142e−06	5.474e−03	1.000e+00
TGFBRAP1	4.709144	7.462790e−06	5.474e−03	1.000e+00
MOV10L1	4.725343	6.891827e−06	5.474e−03	1.000e+00
GJB4	4.664166	9.296134e−06	5.809e−03	1.000e+00
CATSPERB	−4.616693	1.169712e−05	6.537e−03	1.000e+00

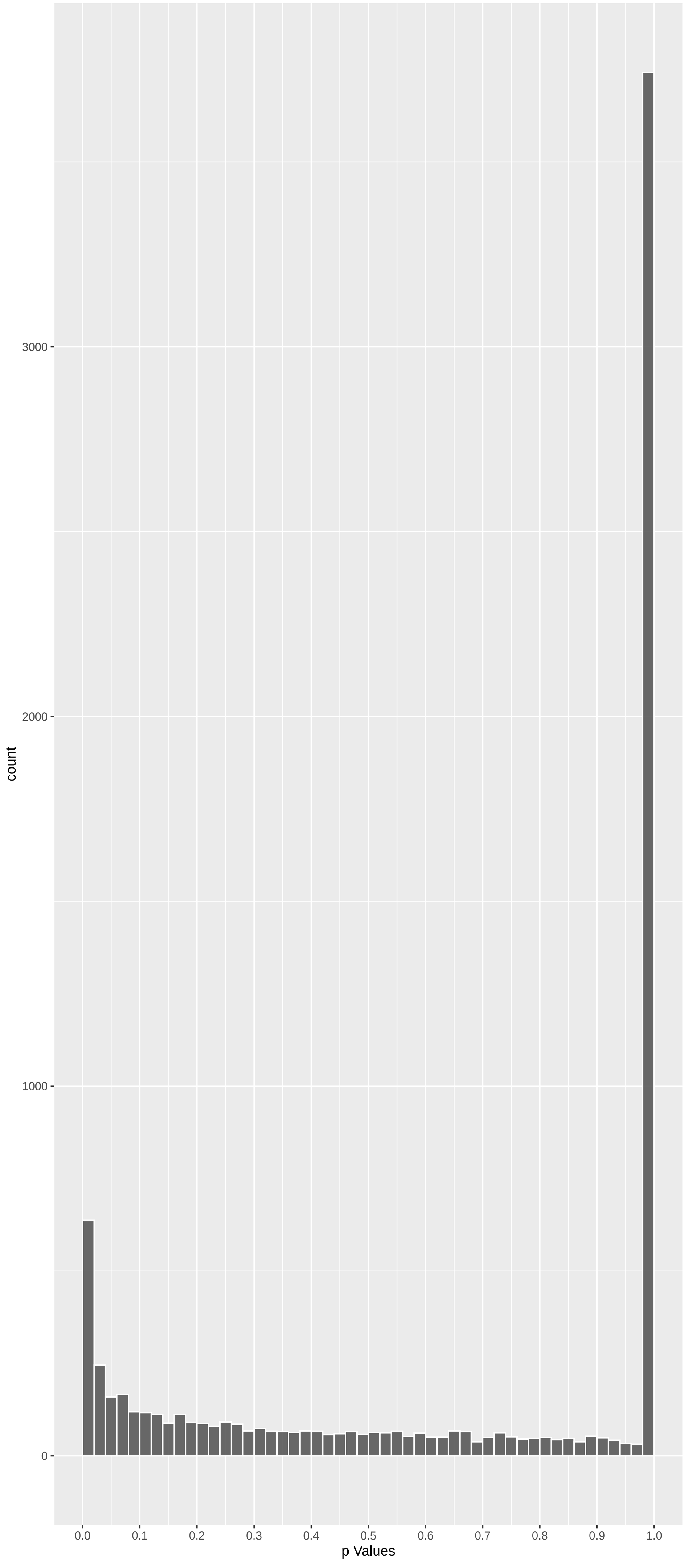
Top genes by Q−Value non−permulated

Gene	Rho	P	p.adj	qValueNoperm
M6PR	1.78273750	0.22388713	9.137e−01	1.000e+00
FKBP4	1.72944567	0.25118809	9.704e−01	1.000e+00
CYP26B1	−0.74321024	1.00000000	1.000e+00	1.000e+00
NDUFAF7	−2.14121206	0.09677082	5.876e−01	1.000e+00
FUCA2	−1.87925889	0.18062746	8.181e−01	1.000e+00
DBNDD1	1.66415418	0.28824487	1.000e+00	1.000e+00
HS3ST1	0.21161908	1.00000000	1.000e+00	1.000e+00
SEMA3F	−1.26578439	0.61677086	1.000e+00	1.000e+00
CFTR	0.93054981	1.00000000	1.000e+00	1.000e+00
CYP51A1	0.64636297	1.00000000	1.000e+00	1.000e+00
USP28	1.51972845	0.38573771	1.000e+00	1.000e+00
SLC7A2	−0.22117348	1.00000000	1.000e+00	1.000e+00
HSPB6	0.05250693	1.00000000	1.000e+00	1.000e+00
PDK4	2.56154456	0.03126236	3.236e−01	1.000e+00
USH1C	0.76624172	1.00000000	1.000e+00	1.000e+00
BAIAP2L1	1.40740597	0.47792113	1.000e+00	1.000e+00
TMEM132A	1.38398504	0.49908911	1.000e+00	1.000e+00
DVL2	−1.61344257	0.31994517	1.000e+00	1.000e+00
RPAP3	−0.95929364	1.00000000	1.000e+00	1.000e+00
HOXA11	0.40665731	1.00000000	1.000e+00	1.000e+00
TRAPPC6A	−0.02982768	1.00000000	1.000e+00	1.000e+00
SPATA20	−1.03102890	0.90758181	1.000e+00	1.000e+00
CD79B	−0.72410073	1.00000000	1.000e+00	1.000e+00
RHBDD2	0.88259060	1.00000000	1.000e+00	1.000e+00
RPUSD1	1.27002697	0.61222507	1.000e+00	1.000e+00

Positive Rho Non−permulated



Negative Rho Non−permulated



Top Positive genes by P−value non−permulated

Gene	Rho	P	p.adj	qValueNoperm
FBXO24	5.743811	2.777080e−08	2.082e−04	1.000e+00
PIWIL1	5.370477	2.355855e−07	7.066e−04	1.000e+00
TRANK1	5.180663	6.632962e−07	1.338e−03	1.000e+00
MYH13	4.764435	5.681512e−06	5.325e−03	1.000e+00
LPIN1	4.776008	5.364286e−06	5.325e−03	1.000e+00
KCNQ3	4.685089	8.395142e−06	5.474e−03	1.000e+00
TGFBRAP1	4.709144	7.462790e−06	5.474e−03	1.000e+00
MOV10L1	4.725343	6.891827e−06	5.474e−03	1.000e+00
GJB4	4.664166	9.296134e−06	5.809e−03	1.000e+00
PARP12	4.608373	1.217497e−05	6.537e−03	1.000e+00

Top Negative genes by P−value non−permulated

Gene	Rho	P	p.adj	qValueNoperm
LRPPRC	−5.814897	1.820133e−08	2.082e−04	1.000e+00
METAP1D	−5.568879	7.691524e−08	3.845e−04	1.000e+00
CLCNKA	−5.462595	1.407671e−07	5.278e−04	1.000e+00
CATSPERE	−5.203486	5.867539e−07	1.338e−03	1.000e+00
TET1	−5.155633	7.583248e−07	1.338e−03	1.000e+00
THADA	−5.144860	8.031603e−07	1.338e−03	1.000e+00
KIF18B	−5.040409	1.393614e−06	2.090e−03	1.000e+00
TMEM126A	−4.996415	1.752173e−06	2.190e−03	1.000e+00
CENPT	−5.008843	1.642747e−06	2.190e−03	1.000e+00
SLFN14	−4.891836	2.996988e−06	3.457e−03	1.000e+00

GO_Biological_Process_2023 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
NADH Dehydrogenase Complex Assembly (GO:	−0.31854752	48	2.365e−14	6.367e−11	TMEM126A:9 NDUFS3:34 NDUFS5:92 NDUFA9:94 NDUFB8:99 NDUFB6:144
Mitochondrial Respiratory Chain Complex	−0.31854752	48	2.365e−14	6.367e−11	TMEM126A:9 NDUFS3:34 NDUFS5:92 NDUFA9:94 NDUFB8:99 NDUFB6:144
Mitochondrial Respiratory Chain Complex	−0.22831413	76	6.325e−12	1.135e−08	TMEM126A:9 NDUFS3:34 NDUFS5:92 NDUFA9:94 NDUFB8:99 NDUFB6:144
Monoatomic Cation Transmembrane Transpor	0.12226080	265	9.404e−12	1.266e−08	KCNQ3:8 KCNV2:43 TRPM5:73 KCNF1:147 KCNA10:167 SCNN1G:183
Mitochondrial Gene Expression (GO:014005	−0.18121894	99	4.994e−10	5.379e−07	POLRMT:30 TFB2M:45 PTCD3:108 MRPL37:110 FASTKD1:147 FASTKD2:380
tRNA Methylation (GO:0030488)	−0.29393566	36	1.063e−09	9.542e−07	THADA:6 TRMT10C:27 THUMPD2:31 TRMT10B:232 TRMO:248 TRMT10A:258
Inorganic Cation Transmembrane Transport	0.10722450	268	1.854e−09	1.426e−06	KCNQ3:8 KCNV2:43 TRPM5:73 KCNF1:147 KCNA10:167 SCNN1G:183
Potassium Ion Transmembrane Transport (G	0.15197531	129	2.748e−09	1.850e−06	KCNQ3:8 KCNV2:43 KCNF1:147 KCNA10:167 KCNS3:209 KCNK18:275
Oxidative Phosphorylation (GO:0006119)	−0.23087484	54	4.539e−09	2.716e−06	NDUFS3:34 NDUFS5:92 NDUFA9:94 NDUFB8:99 NDUFB6:144 ATP5PB:240
Chemical Synaptic Transmission (GO:00072	0.11023545	234	7.371e−09	3.969e−06	KCNQ3:8 RIMBP2:25 CACNB2:52 HRH3:78 GABRB3:111 DRD2:213
Proton Motive Force−Driven Mitochondrial	−0.23854919	47	1.572e−08	7.694e−06	NDUFS3:34 NDUFS5:92 NDUFA9:94 NDUFB8:99 NDUFB6:144 ATP5PB:240
Mitochondrial Electron Transport, NADH T	−0.28594444	31	3.639e−08	1.447e−05	NDUFS3:34 NDUFS5:92 NDUFA9:94 NDUFB8:99 NDUFB6:144 NDUFB10:271
Modulation Of Chemical Synaptic Transmis	0.15719355	103	3.761e−08	1.447e−05	TRIO:114 CACNG5:152 CPLX3:158 DRD2:213 DLGAP2:222 OTOF:261
Proton Motive Force−Driven ATP Synthesis	−0.22396591	51	3.235e−08	1.447e−05	NDUFS3:34 NDUFS5:92 NDUFA9:94 NDUFB8:99 NDUFB6:144 ATP5PB:240
Aerobic Respiration (GO:0009060)	−0.21858963	52	5.096e−08	1.759e−05	NDUFS3:34 NDUFS5:92 NDUFA9:94 NDUFB8:99 NDUFB6:144 NDUFB10:271
tRNA Modification (GO:0006400)	−0.19398202	66	5.227e−08	1.759e−05	THADA:6 THUMPD2:31 DTWD1:215 TRMO:248 TRMT10A:258 FTSJ1:318
Potassium Ion Transport (GO:0006813)	0.14504949	115	8.255e−08	2.615e−05	KCNQ3:8 KCNV2:43 KCNF1:147 KCNA10:167 KCNS3:209 KCNK18:275
Mitochondrial Translation (GO:0032543)	−0.15838148	95	1.007e−07	3.012e−05	PTCD3:108 MRPL37:110 FASTKD2:380 MRPS23:465 MTIF2:476 MRPL42:484
Calcium Ion Transport (GO:0006816)	0.15041269	101	1.853e−07	4.990e−05	CRACR2B:46 CACNB2:52 TRPM5:73 ITPR1:250 ATP2A2:253 TRPM3:311
Mitochondrial ATP Synthesis Coupled Elec	−0.20948974	52	1.778e−07	4.990e−05	NDUFS3:34 NDUFS5:92 NDUFA9:94 NDUFB8:99 NDUFB6:144 NDUFB10:271
Metal Ion Transport (GO:0030001)	0.12506045	145	2.173e−07	5.319e−05	SLC17A4:115 KCNF1:147 KCNA10:167 SCNN1G:183 KCNS3:209 ITPR1:250
Regulation Of Synaptic Transmission, Glu	0.22121363	46	2.139e−07	5.319e−05	CACNG5:152 SHANK3:184 DRD2:213 GRM1:337 DGKI:387 CACNG4:514
Regulation Of DNA−templated Transcriptio	−0.04094609	1447	4.435e−07	1.038e−04	ZSCAN20:28 BRCA1:66 ZSCAN5C:101 ATP2B4:107 ZFP37:113 ZMYND11:120
Aerobic Electron Transport Chain (GO:001	−0.20023570	52	6.022e−07	1.351e−04	NDUFS3:34 NDUFS5:92 NDUFA9:94 NDUFB8:99 NDUFB6:144 NDUFB10:271
Anterograde Trans−Synaptic Signaling (GO	0.11101869	168	7.480e−07	1.611e−04	KCNQ3:8 CACNB2:52 HRH3:78 GABRB3:111 GLRA3:239 HTR1B:240
Cardiac Conduction (GO:0061337)	0.22169795	40	1.243e−06	2.575e−04	CACNB2:52 DSC2:225 KCNJ5:390 BIN1:456 DSG2:537 DSP:649
Mitochondrial RNA Metabolic Process (GO:	−0.31003325	19	2.906e−06	5.795e−04	POLRMT:30 TFB2M:45 PNPT1:137 FASTKD1:147 FASTKD2:380 TEFM:452
piRNA Processing (GO:0034587)	0.32622871	17	3.226e−06	6.079e−04	PIWIL1:2 MOV10L1:6 ASZ1:63 GPAT2:71 FKBP6:116 DDX4:127
Translation (GO:0006412)	−0.10289002	173	3.274e−06	6.079e−04	NARS1:76 PTCD3:108 MRPL37:110 FTSJ1:318 RARS1:325 ZAR1:329
Inorganic Cation Import Across Plasma Me	0.13770854	93	4.610e−06	8.275e−04	SCNN1G:183 CACNA1B:282 KCNJ5:390 CACNA1I:413 TRPV2:468 TRPM1:479
Regulation Of Heart Rate By Cardiac Cond	0.22162099	35	5.764e−06	1.001e−03	CACNB2:52 DSC2:225 KCNJ5:390 BIN1:456 DSG2:537 DSP:649
Cell Junction Assembly (GO:0034329)	0.14489373	80	7.686e−06	1.293e−03	SDK2:26 DBNL:69 SHANK2:100 SHANK3:184 DRD2:213 ITGA6:235
RNA Methylation (GO:0001510)	−0.16807659	57	1.161e−05	1.892e−03	THADA:6 TRMT10C:27 THUMPD2:31 TFB2M:45 TRMO:248 TRMT10A:258
Sensory Perception Of Light Stimulus (GO	0.13358603	90	1.226e−05	1.892e−03	CRYGA:105 WFS1:173 GPR143:256 CHRNB2:427 CRYBB1:478 TRPM1:479
Visual Perception (GO:0007601)	0.13430828	89	1.230e−05	1.892e−03	CRYGA:105 WFS1:173 GPR143:256 CHRNB2:427 CRYBB1:478 TRPM1:479
Cell−Cell Junction Organization (GO:0045	0.15674747	64	1.479e−05	2.212e−03	PKP3:288 GJC1:400 CDH4:531 DSG2:537 DSP:649 PKP1:716
Cellular Respiration (GO:0045333)	−0.15179197	68	1.538e−05	2.238e−03	NDUFS3:34 NDUFS5:92 NDUFA9:94 NDUFB8:99 NDUFB6:144 NDUFB10:271
Regulation Of Transcription By RNA Polym	−0.03432493	1500	1.720e−05	2.438e−03	TET1:5 ZSCAN20:28 BRCA1:66 TLE6:100 ZSCAN5C:101 ATP2B4:107
Muscle Contraction (GO:0006936)	0.13628952	83	1.822e−05	2.515e−03	MYH13:5 MYH2:19 CKMT2:88 EMD:160 MYOM2:161 MYOF:271
Neuron Projection Morphogenesis (GO:0048	0.11253550	121	1.993e−05	2.683e−03	DBNL:69 TRIO:114 SHANK3:184 CNTN4:198 DRD2:213 DIP2A:219

EnrichmentHsSymbolsFile2 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
REACTOME_NEURONAL_SYSTEM	0.13583090	345	5.828e−18	3.780e−14	KCNQ3:8 KCNV2:43 CACNB2:52 DBNL:69 ADCY1:94 SHANK2:100
WP_MITOCHONDRIAL_COMPLEX_I_ASSEMBLY_MODE	−0.31371317	47	1.022e−13	3.314e−10	NDUFS3:34 NDUFS5:92 NDUFB8:99 NDUFB6:144 ECSIT:245 TMEM186:257
REACTOME_COMPLEX_I_BIOGENESIS	−0.30113481	48	5.388e−13	1.165e−09	NDUFS3:34 NDUFS5:92 NDUFA9:94 NDUFB8:99 NDUFB6:144 ECSIT:245
FISCHER_DREAM_TARGETS	−0.07366194	802	2.134e−12	3.460e−09	KIF18B:7 MIS18BP1:17 SMC4:49 FAM161A:51 FANCM:56 BRCA1:66
REACTOME_RESPIRATORY_ELECTRON_TRANSPORT	−0.21705506	79	2.651e−11	3.439e−08	LRPPRC:1 NDUFS3:34 NDUFS5:92 NDUFA9:94 NDUFB8:99 NDUFB6:144
REACTOME_KERATINIZATION	0.20984294	83	4.015e−11	4.341e−08	DSG3:103 KLK13:176 KRT80:195 DSC2:225 KRT17:255 PKP3:288
KEGG_NEUROACTIVE_LIGAND_RECEPTOR_INTERAC	0.11967217	255	5.316e−11	4.926e−08	SSTR4:57 HRH3:78 P2RX2:86 GABRB3:111 P2RY4:186 DRD2:213
REACTOME_RESPIRATORY_ELECTRON_TRANSPORT_	−0.19105521	98	6.593e−11	5.346e−08	LRPPRC:1 NDUFS3:34 NDUFS5:92 NDUFA9:94 NDUFB8:99 NDUFB6:144
NIKOLSKY_BREAST_CANCER_7P22_AMPLICON	0.30113335	38	1.346e−10	9.702e−08	TTYH3:68 BRAT1:464 AP5Z1:519 SNX8:552 RADIL:594 INTS1:635
MIKKELSEN_MEF_HCP_WITH_H3K27ME3	0.08205396	508	3.090e−10	2.005e−07	KCNQ3:8 CRACR2B:46 NETO1:50 DKKL1:75 ALOX5:79 BOLL:82
BENPORATH_ES_WITH_H3K27ME3	0.06242213	898	3.433e−10	2.025e−07	KCNQ3:8 FRMPD4:45 SMPD3:51 NKPD1:53 C1orf94:56 PLEC:101
WP_GPCRS_CLASS_A_RHODOPSINLIKE	0.12732830	203	4.363e−10	2.358e−07	SSTR4:57 HRH3:78 OPN3:91 P2RY4:186 DRD2:213 CCR4:228
REACTOME_SENSORY_PERCEPTION	0.10902447	277	4.792e−10	2.391e−07	CACNB2:52 TRPM5:73 PDE6B:121 OR10D3:123 GNAT3:164 SCNN1G:183
JOHNSTONE_PARVB_TARGETS_3_DN	−0.06948614	689	6.932e−10	3.212e−07	MIS18BP1:17 N4BP2:29 SMC4:49 CCPG1:58 NDUFS5:92 QSER1:95
REACTOME_FORMATION_OF_THE_CORNIFIED_ENVE	0.20233458	76	1.099e−09	4.754e−07	DSG3:103 KLK13:176 KRT80:195 DSC2:225 KRT17:255 PKP3:288
YOSHIMURA_MAPK8_TARGETS_UP	0.05562861	1052	1.730e−09	7.014e−07	KCNQ3:8 AFDN:12 SSTR4:57 CASP2:58 RNF112:65 ALOX5:79
KEGG_CALCIUM_SIGNALING_PATHWAY	0.13652673	161	2.406e−09	9.183e−07	P2RX2:86 ADCY1:94 PHKG1:197 ADRB1:244 ITPR1:250 ATP2A2:253
REACTOME_SIGNALING_BY_GPCR	0.07176626	596	2.778e−09	1.001e−06	SSTR4:57 HRH3:78 OPN3:91 ADCY1:94 TRIO:114 PIK3R5:134
REACTOME_MUSCLE_CONTRACTION	0.13054939	172	3.727e−09	1.272e−06	NEB:27 MYBPC2:36 CACNB2:52 TNNI2:125 ITPR1:250 ATP2A2:253
MARSON_BOUND_BY_E2F4_UNSTIMULATED	−0.07073743	602	3.928e−09	1.274e−06	KIF18B:7 ETAA1:33 SMC4:49 FANCM:56 HROB:72 ORC1:74
REACTOME_THE_CITRIC_ACID_TCA_CYCLE_AND_R	−0.13993940	145	6.316e−09	1.951e−06	LRPPRC:1 NDUFS3:34 NDUFS5:92 NDUFA9:94 NDUFB8:99 NDUFB6:144
WP_OXIDATIVE_PHOSPHORYLATION	−0.24698494	46	6.893e−09	2.032e−06	NDUFS3:34 NDUFS5:92 NDUFA9:94 NDUFB8:99 NDUFB6:144 ATP5PB:240
REACTOME_VOLTAGE_GATED_POTASSIUM_CHANNEL	0.25423681	41	1.790e−08	5.049e−06	KCNQ3:8 KCNV2:43 KCNF1:147 KCNA10:167 KCNS3:209 KCNH3:299
NIKOLSKY_BREAST_CANCER_20Q12_Q13_AMPLICO	0.14843840	118	2.652e−08	7.169e−06	HRH3:78 SYCP2:151 TCFL5:345 FAM217B:414 ZBTB46:445 CABLES2:525
MIKKELSEN_MCV6_HCP_WITH_H3K27ME3	0.08165083	390	3.574e−08	9.274e−06	KCNQ3:8 SDK2:26 FAM83G:80 IRX4:136 TDRD6:191 DLGAP2:222
MARTENS_TRETINOIN_RESPONSE_UP	0.06371599	641	4.648e−08	1.160e−05	GJB4:9 KIF2B:16 UBQLN3:34 C16orf90:37 HRH3:78 ZC3H3:90
DODD_NASOPHARYNGEAL_CARCINOMA_DN	−0.04916409	1109	4.969e−08	1.194e−05	LRPPRC:1 TET1:5 KIF18B:7 THUMPD2:31 TFB2M:45 SMC4:49
REACTOME_POTASSIUM_CHANNELS	0.17186718	84	5.306e−08	1.229e−05	KCNQ3:8 KCNV2:43 KCNF1:147 KCNA10:167 KCNS3:209 KCNK18:275
REACTOME_TRANSMISSION_ACROSS_CHEMICAL_SY	0.10576952	210	1.355e−07	3.032e−05	CACNB2:52 ADCY1:94 GABRB3:111 GNAT3:164 AP2A2:232 GLRA3:239
RODRIGUES_THYROID_CARCINOMA_POORLY_DIFFE	−0.06672362	529	1.798e−07	3.888e−05	LRPPRC:1 SMC4:49 CCPG1:58 BRCA1:66 NEDD4:98 MRPL37:110
MEISSNER_NPC_HCP_WITH_H3K4ME2_AND_H3K27M	0.08631997	305	2.375e−07	4.969e−05	HRH3:78 FAM163B:135 CADM3:143 TMEM151A:205 KCNS3:209 DLGAP2:222
WP_CALCIUM_REGULATION_IN_CARDIAC_CELLS	0.13819431	115	3.165e−07	6.417e−05	GJB4:9 ADCY1:94 RGS9:188 ADRB1:244 ITPR1:250 ATP2A2:253
REACTOME_INTEGRATION_OF_ENERGY_METABOLIS	0.16610491	79	3.391e−07	6.666e−05	CACNB2:52 ADCY1:94 KCNS3:209 ITPR1:250 KCNC2:436 PRKCA:440
REACTOME_GPCR_LIGAND_BINDING	0.07500770	385	4.888e−07	9.325e−05	SSTR4:57 HRH3:78 OPN3:91 P2RY4:186 DRD2:213 CCR4:228
REACTOME_MITOCHONDRIAL_TRANSLATION	−0.15149083	91	6.028e−07	1.117e−04	PTCD3:108 MRPL37:110 MRPS35:116 MRPS23:465 MTIF2:476 MRPL42:484
REACTOME_G_ALPHA_I_SIGNALLING_EVENTS	0.09205190	245	7.455e−07	1.343e−04	SSTR4:57 OPN3:91 ADCY1:94 GNAT3:164 P2RY4:186 RGS9:188
GEORGES_TARGETS_OF_MIR192_AND_MIR215	−0.05341443	751	7.860e−07	1.378e−04	DST:15 MIS18BP1:17 LIMCH1:19 N4BP2:29 SMC4:49 FANCM:56
NIKOLSKY_BREAST_CANCER_12Q24_AMPLICON	0.36773311	15	8.183e−07	1.397e−04	P2RX2:86 EP400:87 GOLGA3:391 GALNT9:412 SFSWAP:495 POLE:851
REACTOME_CLASS_A_1_RHODOPSIN_LIKE_RECEPT	0.08351310	293	9.486e−07	1.578e−04	SSTR4:57 HRH3:78 OPN3:91 P2RY4:186 DRD2:213 CCR4:228
WP_ELECTRON_TRANSPORT_CHAIN_OXPHOS_SYSTE	−0.16593757	71	1.354e−06	2.196e−04	NDUFS3:34 NDUFS5:92 NDUFA9:94 NDUFB8:99 NDUFB6:144 ATP5PB:240

DisGeNET Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Mitochondrial Diseases	−0.11573909	333	5.447e−13	5.342e−09	LRPPRC:1 TMEM126A:9 TRMT10C:27 NDUFS3:34 SARS2:82 NDUFS5:92
Increased CSF lactate	−0.21029171	52	1.593e−07	7.811e−04	LRPPRC:1 TRMT10C:27 NDUFS3:34 NDUFA9:94 NDUFA11:272 TMEM126B:278
Schizophrenia	0.03993086	1503	5.307e−07	1.735e−03	TRANK1:3 KCNQ3:8 ASAP1:14 MROH2A:17 NKAPL:24 EPN2:47
Autism Spectrum Disorders	0.06680091	451	1.430e−06	3.506e−03	CACNB2:52 SSTR4:57 SHANK2:100 GABRB3:111 TRIO:114 SYNE1:159
Color vision defect	0.20451491	41	5.943e−06	1.151e−02	OFD1:40 SEMA4A:166 TTLL5:233 CDHR1:452 PRPH2:613 CACNA2D4:664
Congenital cataract	0.13798573	89	7.041e−06	1.151e−02	CRYGA:105 WFS1:173 RAB3GAP2:201 ATF4:432 CRYAA2:477 CRYBB1:478
Color vision defect, severe	0.21023990	36	1.283e−05	1.260e−02	OFD1:40 SEMA4A:166 TTLL5:233 CDHR1:452 PRPH2:613 CACNA2D4:664
Abnormal color vision	0.21023990	36	1.283e−05	1.260e−02	OFD1:40 SEMA4A:166 TTLL5:233 CDHR1:452 PRPH2:613 CACNA2D4:664
Loss in color vision	0.21023990	36	1.283e−05	1.260e−02	OFD1:40 SEMA4A:166 TTLL5:233 CDHR1:452 PRPH2:613 CACNA2D4:664
NADH:Q(1) Oxidoreductase deficiency	−0.25215086	25	1.285e−05	1.260e−02	NDUFS3:34 NDUFB10:271 NDUFA11:272 TMEM126B:278 NDUFAF1:382 NDUFAF4:390
Difficulties with night vision	0.11886660	109	1.876e−05	1.672e−02	KCNV2:43 PDE6B:121 SEMA4A:166 TTLL5:233 CDHR1:452 TRPM1:479
Acute necrotizing encephalopathy	−0.27694047	19	2.938e−05	2.216e−02	NDUFS3:34 NDUFA11:272 TMEM126B:278 NDUFAF1:382 NDUFAF4:390 TIMMDC1:457
Lactic acidemia	−0.11162049	118	2.923e−05	2.216e−02	LRPPRC:1 TRMT10C:27 NDUFS3:34 NDUFA9:94 SFXN4:170 NDUFA11:272
Night Blindness	0.11200710	114	3.743e−05	2.621e−02	KCNV2:43 PDE6B:121 SEMA4A:166 TTLL5:233 CDHR1:452 TRPM1:479
CATARACT, COPPOCK−LIKE	0.39509678	9	4.056e−05	2.652e−02	CRYBB1:478 GJA8:543 BFSP2:581 GJA3:921 CRYBB2:1806 VIM:1844
MITOCHONDRIAL COMPLEX I DEFICIENCY	−0.23066791	26	4.705e−05	2.884e−02	NDUFS3:34 NDUFA11:272 TMEM126B:278 NDUFAF1:382 NDUFAF4:390 TIMMDC1:457
Psychoses, Substance−Induced	0.30165720	15	5.245e−05	3.008e−02	RGS9:188 CAMK2B:1295 AKT1:1420 SLC6A3:1451 PICK1:1695 ESR1:1748
Mitochondrial Respiratory Chain Deficien	−0.19705510	35	5.522e−05	3.008e−02	PNPT1:137 HSD17B6:479 RARS2:599 PPRC1:620 ROS1:710 FOXRED1:748
Psychoses, Drug	0.26233965	19	7.560e−05	3.902e−02	RGS9:188 DTNBP1:1146 CAMK2B:1295 AKT1:1420 SLC6A3:1451 PICK1:1695
Huntington Disease	0.05207544	496	8.376e−05	4.107e−02	SH3BP2:30 SORL1:48 SSTR4:57 CASP2:58 ALOX5:79 DRD2:213
Abnormal mitochondria in muscle tissue	−0.22604616	25	9.192e−05	4.292e−02	NDUFS3:34 NDUFB10:271 NDUFA11:272 TMEM126B:278 NDUFAF1:382 NDUFAF4:390
Acidosis, Lactic	−0.09229570	145	1.300e−04	4.553e−02	LRPPRC:1 TRMT10C:27 NDUFS3:34 NDUFA9:94 SFXN4:170 NDUFA11:272
Night blindness, congenital stationary	0.23064200	23	1.293e−04	4.553e−02	USP11:85 PDE6B:121 TRPM1:479 CACNA2D4:664 GRM6:760 GPR34:904
Channelopathies	0.16801221	44	1.165e−04	4.553e−02	SHANK3:184 KCNJ5:390 CACNA1D:914 SCN5A:994 KCNJ3:1129 KCNA4:1176
Progressive macrocephaly	−0.23660039	22	1.227e−04	4.553e−02	NDUFS3:34 NDUFA11:272 TMEM126B:278 NDUFAF1:382 NDUFAF4:390 TIMMDC1:457
Severe myopia	0.11952071	88	1.091e−04	4.553e−02	PDE6B:121 ATP6V0A2:281 ATF4:432 TRPM1:479 CACNA2D4:664 ZIC2:737
Substance abuse problem	0.09623584	134	1.244e−04	4.553e−02	PIWIL1:2 GABRB3:111 SGSM3:130 DOCK1:157 SYNE1:159 WDR11:171
Sudden Cardiac Death	0.16461331	46	1.134e−04	4.553e−02	CACNB2:52 EMD:160 DSC2:225 MYL3:285 DSG2:537 DSP:649
Cataract, Pulverulent	0.36751387	9	1.347e−04	4.554e−02	CRYBB1:478 GJA8:543 BFSP2:581 GJA3:921 CRYBB2:1806 VIM:1844
Abnormality of macular pigmentation	0.23478913	21	1.963e−04	6.095e−02	PDE6B:121 TRPM1:479 PRPH2:613 CACNA2D4:664 GRM6:760 ELOVL4:825
Bipolar Disorder	0.04260830	671	2.035e−04	6.095e−02	TRANK1:3 KCNQ3:8 SORL1:48 CACNB2:52 SSTR4:57 DBNL:69
Macular pigmentary changes	0.23478913	21	1.963e−04	6.095e−02	PDE6B:121 TRPM1:479 PRPH2:613 CACNA2D4:664 GRM6:760 ELOVL4:825
Nuclear non−senile cataract	0.19926817	29	2.051e−04	6.095e−02	WFS1:173 CRYBB1:478 GJA8:543 CRYAA:673 CRYBB3:886 GJA3:921
Autistic Disorder	0.04477948	576	2.811e−04	7.984e−02	FRMPD4:45 NETO1:50 CACNB2:52 SHANK2:100 GABRB3:111 TMEM132B:174
Growth retardation	−0.08688568	146	3.011e−04	7.984e−02	MAN2B1:13 DST:15 NDUFS3:34 FANCM:56 PNPT1:137 NPHS1:196
Pfaundler−Hurler Syndrome	−0.21319195	24	3.013e−04	7.984e−02	MAN2B1:13 VCAN:87 GNPTAB:321 ARSB:464 NAGLU:675 GUSB:811
Poor growth	−0.08688568	146	3.011e−04	7.984e−02	MAN2B1:13 DST:15 NDUFS3:34 FANCM:56 PNPT1:137 NPHS1:196
Nuclear cataract	0.18720716	31	3.111e−04	8.027e−02	WFS1:173 CRYBB1:478 GJA8:543 CRYAA:673 CRYBB3:886 GJA3:921
Growth delay	−0.08622058	147	3.196e−04	8.037e−02	MAN2B1:13 DST:15 NDUFS3:34 FANCM:56 PNPT1:137 NPHS1:196
Electroretinogram abnormal	0.09283442	123	3.880e−04	9.513e−02	OFD1:40 PDE6B:121 SEMA4A:166 LZTFL1:381 CDHR1:452 TRPM1:479

MGI_Mammalian_Phenotype_Level_4 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
MP0003635 abnormal synaptic transmissio	0.08281771	381	7.372e−08	3.509e−05	IQSEC1:33 NETO1:50 HRH3:78 ARC:84 ADCY1:94 GABRB3:111
MP0005551 abnormal eye electrophysiolog	0.10983361	134	1.396e−05	3.322e−03	CACNB2:52 PDE6B:121 SV2B:142 CPLX3:158 SEMA4A:166 RGS9:188
MP0002085 abnormal embryonic tissue	−0.05265556	583	3.484e−05	5.529e−03	FUZ:22 CHRD:40 BRCA1:66 TRPM6:67 NSD1:124 ADAMTS20:153
MP0002086 abnormal extraembryonic tissu	−0.05826971	429	6.411e−05	6.103e−03	CHRD:40 PTPRB:55 BRCA1:66 NSD1:124 RLIM:149 DLGAP5:192
MP0002429 abnormal blood cell	−0.03942756	1124	5.540e−05	6.103e−03	MAN2B1:13 PTPN13:60 BRCA1:66 NEDD4:98 POLQ:131 JAK3:142
MP0002572 abnormal emotion/affect behav	0.07039395	273	8.957e−05	7.106e−03	PLA2G6:77 HRH3:78 ARC:84 ADCY1:94 GABRB3:111 WFS1:173
MP0002066 abnormal motor capabilities/c	0.03963142	982	1.150e−04	7.821e−03	LPIN1:4 NEB:27 DBNL:69 PLA2G6:77 HRH3:78 KIF1A:96
MP0002971 abnormal brown adipose	0.13264024	70	1.351e−04	8.040e−03	LPIN1:4 ADRB1:244 GRM1:337 AGPAT2:472 GPR68:553 RPTOR:638
MP0005253 abnormal eye physiology	0.11193102	87	3.375e−04	1.344e−02	CACNB2:52 SEMA4A:166 SORT1:199 GLRA3:239 GJC1:400 TRPM1:479
MP0000716 abnormal immune system	−0.03788023	917	3.298e−04	1.344e−02	MAN2B1:13 PTPN13:60 BRCA1:66 NEDD4:98 POLQ:131 JAK3:142
MP0002067 abnormal sensory capabilities	0.06885800	237	3.387e−04	1.344e−02	LPIN1:4 DBNL:69 GABRB3:111 GLRA3:239 ITPR1:250 ATP2A2:253
MP0002064 seizures	0.07047796	230	2.990e−04	1.344e−02	KCNQ3:8 NETO1:50 DBNL:69 GABRB3:111 SV2B:142 DRD2:213
MP0002398 abnormal bone marrow	−0.04063861	743	3.993e−04	1.462e−02	DST:15 BRCA1:66 NEDD4:98 POLQ:131 JAK3:142 AFF1:151
MP0000689 abnormal spleen morphology	−0.04678284	506	5.486e−04	1.742e−02	MAN2B1:13 PKHD1:25 JAK3:142 DNMT1:224 SIGLEC1:279 NFKB2:281
MP0001970 abnormal pain threshold	0.08458093	142	5.791e−04	1.742e−02	SSTR4:57 ADCY1:94 KIF1A:96 PIRT:175 CACNA1B:282 ADORA1:304
MP0001697 abnormal embryo size	−0.05409987	363	5.854e−04	1.742e−02	CHRD:40 PTPRB:55 BRCA1:66 VCAN:87 ECSIT:245 RRN3:276
MP0002063 abnormal learning/memory/cond	0.05640510	325	6.596e−04	1.847e−02	NETO1:50 PLA2G6:77 HRH3:78 ARC:84 ADCY1:94 WFS1:173
MP0003935 abnormal craniofacial develop	−0.06770459	216	7.479e−04	1.978e−02	FUZ:22 CHRD:40 TRPM6:67 HESX1:172 SUPV3L1:337 PAX3:415
MP0005452 abnormal adipose tissue	0.05565087	292	1.392e−03	3.487e−02	LPIN1:4 NEB:27 HRH3:78 DSG3:103 ADRB1:244 PHEX:260
MP0000598 abnormal liver morphology	−0.04558479	440	1.561e−03	3.710e−02	MAN2B1:13 PKHD1:25 ANGPTL3:106 CYBRD1:109 LTA:303 GHR:348
MP0005380 embryogenesis phenotype	−0.04775063	367	2.288e−03	3.710e−02	NACA:32 PTPRB:55 BRCA1:66 RLIM:149 NLRP5:197 PADI6:202
MP0002735 abnormal chemical nociception	0.16739411	28	2.217e−03	3.710e−02	ADCY1:94 PIRT:175 CACNA1B:282 GRIK1:578 OPRL1:656 P2RX3:740
MP0004085 abnormal heartbeat	0.07411310	149	2.025e−03	3.710e−02	CACNB2:52 PTK2:150 SYNE1:159 DRD2:213 ADRB1:244 ATP2A2:253
MP0001486 abnormal startle reflex	0.09474410	90	2.034e−03	3.710e−02	DRD2:213 HTR1B:240 LOXHD1:247 OTOF:261 CACNA1B:282 ADRA2C:467
MP0002084 abnormal developmental patter	−0.05137327	312	2.338e−03	3.710e−02	FUZ:22 CHRD:40 PTPRB:55 BRCA1:66 NSD1:124 DNMT1:224
MP0002080 prenatal lethality	−0.02918487	1198	2.278e−03	3.710e−02	FUZ:22 PKHD1:25 NACA:32 CHRD:40 PTPRB:55 BRCA1:66
MP0003252 abnormal bile duct	−0.19515494	21	1.994e−03	3.710e−02	PKHD1:25 CYP8B1:934 P2RY13:1027 NR1H4:1050 JAG1:1586 MAP3K14:1771
MP0002272 abnormal nervous system	0.07008346	171	1.811e−03	3.710e−02	LPIN1:4 KCNQ3:8 ARC:84 PLEC:101 GABRB3:111 DRD2:213
MP0001905 abnormal dopamine level	0.12624925	49	2.319e−03	3.710e−02	DRD2:213 SLC18A2:300 OPRL1:656 NTSR1:736 CD81:761 NOS3:884
MP0001672 abnormal embryogenesis/ devel	−0.04775063	367	2.288e−03	3.710e−02	NACA:32 PTPRB:55 BRCA1:66 RLIM:149 NLRP5:197 PADI6:202
MP0002135 abnormal kidney morphology	−0.04397644	416	2.931e−03	4.373e−02	MAN2B1:13 PKHD1:25 BRCA1:66 NPHS1:196 TET2:230 HIVEP3:234
MP0002925 abnormal cardiovascular devel	−0.04168575	468	2.940e−03	4.373e−02	FUZ:22 CHRD:40 PTPRB:55 BRCA1:66 VCAN:87 C2CD3:250
MP0002169 no abnormal phenotype	−0.02930197	1075	3.192e−03	4.604e−02	PKHD1:25 BRCA1:66 VCAN:87 INSRR:123 RLIM:149 ADAM9:166
MP0009745 abnormal behavioral response	0.06307795	187	3.374e−03	4.724e−02	GABRB3:111 WFS1:173 DRD2:213 GLRA3:239 SOD1:268 CACNA1B:282
MP0002873 normal phenotype	−0.02872341	1080	3.788e−03	5.009e−02	PKHD1:25 BRCA1:66 VCAN:87 INSRR:123 RLIM:149 ADAM9:166
MP0003119 abnormal digestive system	−0.12135469	48	3.753e−03	5.009e−02	CHRD:40 OVOL2:416 GDF1:663 RUNX1T1:668 GADD45GIP1:887 ERCC5:1000
MP0000432 abnormal head morphology	−0.04896737	285	5.426e−03	6.980e−02	CHRD:40 TRPM6:67 HESX1:172 PAX3:415 TCOF1:418 ARSB:464
MP0002723 abnormal immune serum	−0.03430813	595	6.511e−03	8.156e−02	PTPN13:60 NEDD4:98 POLQ:131 JAK3:142 SERPINA6:212 DGKA:228
MP0002733 abnormal thermal nociception	0.08442524	87	6.856e−03	8.368e−02	PIRT:175 CACNA1B:282 ADORA1:304 L1CAM:639 OPRL1:656 SLC6A2:681
MP0002229 neurodegeneration	0.04930434	244	9.274e−03	1.051e−01	LPIN1:4 CACNB2:52 PLA2G6:77 KIF1A:96 GABRB3:111 PDE6B:121

tissue_specific Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
EBV.lymphocyte	−0.08416947	647	1.744e−12	9.419e−11	KIF18B:7 SLFN14:10 MIS18BP1:17 CERKL:36 SMC4:49 CARMIL2:64
Brodmann.area.9	0.12117417	158	1.847e−07	4.987e−06	KCNQ3:8 FRMPD4:45 NETO1:50 GABRB3:111 DLGAP2:222 GLRA3:239
cerebral.cortex	0.12751645	100	1.164e−05	1.677e−04	KIF1A:96 FAM163B:135 KCNF1:147 ADGRB1:286 KCNH3:299 CHRNB2:427
esophagus.mucosa	0.08783335	213	1.242e−05	1.677e−04	FAM83H:70 DSG3:103 KLK13:176 EPS8L1:190 DSC2:225 BNC1:249
vagina	0.12696948	89	3.759e−05	4.060e−04	DSG3:103 SCNN1G:183 DSC2:225 BNC1:249 SLC18A2:300 NUAK2:369
liver	−0.06714352	320	4.883e−05	4.395e−04	AFM:47 AMBP:90 ANGPTL3:106 AMDHD1:114 SERPINA6:212 PIPOX:226
leg.skin	0.07274540	208	3.463e−04	2.671e−03	GJB4:9 NKPD1:53 FAM83G:80 DSG3:103 PPP1R13L:129 INAVA:133
skeletal.muscle	0.06318101	231	1.079e−03	6.771e−03	MYH2:19 NEB:27 CAVIN4:31 MYBPC2:36 CKMT2:88 TNNI2:125
suprapubic.skin	0.07101877	180	1.129e−03	6.771e−03	GJB4:9 NKPD1:53 DSG3:103 IRX4:136 KRT80:195 BNC1:249
nucleus.accumbens	0.09664831	91	1.514e−03	8.173e−03	INSYN2A:41 OTOF:261 SORCS3:348 JAKMIP1:368 PRKCG:431 CACNG4:514
cerebellar.hemisphere	0.04194731	453	2.813e−03	1.381e−02	SSTR4:57 RNF112:65 ADCY1:94 SV2B:142 CADM3:143 GRM1:337
spleen	−0.04895982	287	4.929e−03	2.218e−02	SLFN14:10 STAB1:65 CPVL:127 STAB2:211 NLRC5:305 FOXP3:443
hypothalamus	0.10145013	57	8.244e−03	3.180e−02	PIRT:175 GLRA3:239 GRIK1:578 PCDH15:648 MC4R:1073 GABRG3:1077
sigmoid.colon	0.10896229	50	7.838e−03	3.180e−02	GJC1:400 TACR2:482 DES:589 NTSR1:736 DPT:1013 CNN1:1084
cerebellum	0.03480995	349	2.825e−02	1.017e−01	TRANK1:3 GLRA4:44 SSTR4:57 RNF112:65 CDR2L:72 HRH3:78
pancreas	−0.06519410	93	3.056e−02	1.031e−01	CATSPERB:16 TMED6:104 FBXW12:163 NPHS1:196 G6PC2:669 SERPINA4:763
putamen	0.15105597	15	4.295e−02	1.364e−01	RGS9:188 PDE10A:624 OPALIN:1019 GPR88:1521 LRRC10B:2253 DIPK1C:2329
uterus	−0.07541285	53	5.819e−02	1.746e−01	ATP2B4:107 HOXD4:474 ADAMTS19:502 PTK7:651 CLEC4F:724 ENPP1:828
subcutaneous.adipose	−0.06482387	70	6.163e−02	1.752e−01	KLB:75 MTARC1:119 LVRN:136 SIGLEC1:279 GHR:348 CFD:500
esophagogastric.junction	−0.17855562	8	8.041e−02	2.147e−01	F2RL2:1552 GADL1:1921 NKX6−1:2550 COL4A5:3607 ADCY5:7734.5 BARX1:7734.5
thyroid	−0.04208959	144	8.348e−02	2.147e−01	CLCNKA:3 TLE6:100 PLA2R1:143 SLC12A8:193 SORD:233 ZNF804B:331
fallopian.tube	−0.06951401	48	9.645e−02	2.367e−01	LDLRAD1:320 ADAMTS3:627 RARRES1:691 CDHR4:855 CCDC198:943 MMP19:1285
bladder	0.09603083	23	1.113e−01	2.612e−01	UGT1A6:333 UPK2:442 PSCA:457 KRT13:518 PADI3:663 PLA2G2F:1161
adrenal.gland	−0.04377159	107	1.197e−01	2.694e−01	CYP17A1:287 SLC16A9:350 KLF14:392 FDXR:429 SLC14A2:430 ADGRV1:609
blood	−0.02656632	266	1.413e−01	3.051e−01	MEFV:93 JAK3:142 EPB42:165 TREML2:186 SHKBP1:378 GATA1:485
coronary.artery	−0.10980086	14	1.551e−01	3.222e−01	SUSD5:1746 MYO18B:2149 MTHFD1L:2284 HEYL:2472 CCN3:7734.5 SPINK13:7734.5
esophagus.muscularis.mucosa	−0.07930416	26	1.621e−01	3.242e−01	PRUNE2:773 JPH2:1150 F2RL2:1552 CCDC69:1772 KCNMB1:1869 GADL1:1921
cortex.kidney	−0.03853033	108	1.688e−01	3.256e−01	PKHD1:25 CLCNKB:41 NPHS1:196 SLC5A12:421 CUBN:683 KIF12:856
amygdala	0.12975997	7	2.346e−01	4.369e−01	PCDH15:648 BCAN:3194 SLC17A7:3484 NEUROD2:7263.5 PTPRZ1:7263.5 SLC38A8:7263.5
breast	0.08635347	14	2.636e−01	4.744e−01	TNN:1504 THRSP:1583 KRT5:2346 ALX4:2686 NPY2R:2780 CIDEC:3437
left.ventricle	0.03595729	67	3.104e−01	5.406e−01	CKMT2:88 MYOM2:161 MYH7B:178 MYL3:285 LMOD2:375 TNNT2:820
aorta	−0.01983223	150	4.054e−01	6.535e−01	VCAN:87 ART4:566 HMCN1:608 OMD:658 FRZB:793 SCRG1:907
atrium.auricle	−0.02494073	89	4.181e−01	6.535e−01	TNNI3K:14 MYPN:302 XIRP1:520 MLIP:528 CCDC141:774 NMRK2:784
ectocervix	−0.04784499	25	4.082e−01	6.535e−01	COL27A1:501 PAPPA2:806 PTHLH:1129 EDNRA:1809 MMRN1:1917 SOCS2:1942
endocervix	0.02795524	69	4.236e−01	6.535e−01	PWP2:217 SLC18A2:300 ADRA2C:467 TMEM132C:588 ELFN1:654 SPON2:770
prostate	0.03448000	38	4.628e−01	6.760e−01	P2RX2:86 PSCA:457 CDH26:781 ACP3:872 NTF4:1643 TRPM8:1795
stomach	0.02338999	83	4.632e−01	6.760e−01	MYH13:5 PSCA:457 CLDN18:788 FUT1:826 TM4SF5:998 SLC9A3:1034
greater.omentum	0.03079656	32	5.472e−01	7.775e−01	BNC1:249 GPR4:815 TIMP4:906 FFAR3:1076 HAS1:1150 THRSP:1583
Brodmann.area.24	0.02635765	28	6.297e−01	7.908e−01	KCNS1:918 CSPG5:925 SHANK1:1015 GRIN3A:1654 DRD5:2186 SLC25A18:2247
C1.spinal.cord	−0.01121010	166	6.212e−01	7.908e−01	SLCO1A2:70 HS3ST5:103 PAQR6:185 STMN4:191 CERS1:473 EVI2A:692